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MILITARY SPECIFICATION

SWITCH, ALARM, LATCHING, SA-1954/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a floor or wall mounted manually operated duress switch which allows an alarm to be initiated by its operator when placed under duress. It is a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Communications-Electronics Command, CECOM RDE Center, ATTN: AMSEL-LC-IEW-D-ED, 10115 Gridley Road Suite 228, Fort Belvoir, VA 22060-5849 by using the self-addressed Standardization Document (Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.
AMSC N/A FSC 6350
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SPECIFICATIONS

FEDERAL

QQ-S-781	-	Strapping, Steel, and Seals.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-B-117	-	Bag, Sleeve and Tubing - Interior Packaging.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-810[1/]	-	Environmental Test Methods.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, With Appropriate Test Methods.
MIL-STD-45662	-	Calibration System Requirements.

[1/] The "C" revision of MIL-STD-810 shall be used.

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

TA13220E3800	-	Switch, Alarm, Latching, SA-1954/FRS-9(V).
TA13220E3030	-	Printed Wiring Board Assembly, LAS.

(Copies of drawings are available from the USA Communications-Electronics Command, CECOM RDE Center, ATTN: AMSEL-LC-IEW-D-CE, 10115 Gridley Road Suite 228, Fort Belvoir, VA 22060-5849.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The latching alarm switch (hereinafter called "LAS" shall be in accordance with top assembly TA13220E380 and as specified herein.

3.1.1 Major components. The LAS shall consist of the following major component:

a. Printed wiring board (PWB) assembly TA13220E3030 (see 3.5.5).

3.1.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the LAS and PWB assemblies when furnished separately. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The LAS shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the LAS may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific materials, material finish or treatment for use with components and sub-components and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the LAS may be newly fabricated from recovered materials to the maximum extent practicable, provided the LAS produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the LAS.

3.4 Environmental conditions. The LAS and PWB assemblies when furnished separately shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.4.3), performance degradation, or malfunction when tested as specified in 4.5.2.6.

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3.4.1 Operational.

a. Temperature range of -20 °F to +150 °F.
b. Humidity up to 95 percent relative.

3.4.2 Nonoperational condition and storage.

a. Temperature range of -30 °F to +165 °F.
b. Humidity up to 95 percent relative.
c. Shock: bench handling.

3.5 Detailed technical requirements.

3.5.1 Tamper protection. The tamper switch shall be mounted in the LAS such that the switch contacts open with not less than 1/16 inch nor more than 1/4 inch of cover movement and before the cover flange is disengaged. The resistance across terminals TB1-5 and -6 shall be less than 1 ohm with the contacts closed (no tamper alarm) and greater than 1 megohm with contacts open (tamper alarm).

3.5.2 Power. The LAS shall function as specified herein on supplied voltages between 19 and 21 Vdc. The required current shall be 20 ±10 miliamperes (mA). Reverse voltage protection shall be incorporated.

3.5.3 LAS operation. Operation of the LAS shall be as follows:

a. After initial power application and with no duress alarm inputs since the last reset, the following non-alarm output conditions shall exist:

1. The resistance across terminals TB1-1 and TB1-2 shall be less than 500 ohms;
2. The light emitting diode (LED) shall not be lit.

b. Momentary activation of the duress switch (S1) shall cause the LAS to latch in the alarm state such that:

1. The resistance across terminals TB1-1 and TB1-2 shall be greater than 100 kilohms (100K ohms);
2. The LED shall be lit.

c. Momentary activation of the reset switch (S2) shall cause return of the LAS to the non-alarm state.

3.5.4 Durability. The LAS shall be sufficiently durable to withstand the test specified in 4.5.2.7 without component failures, inability to pass the functional test (see 4.5.2.5.1), or permanent deformation.

3.5.5 PWB assembly. The PWB assembly shall be in accordance with TA13220E3030 and as specified herein. Operation shall be as follows with 20 ±1 Vdc applied to terminal +20 and the 20 volt (V) return to terminal 3 (ground) and with a LED in accordance with D13220E3816 placed between terminals VDD and 6 (LED anode connected to VDD):

a. After initial power application and with ground potential applied to the alarm input terminal (terminal 5), the following conditions shall exist:

1. Resistance across terminals 2 and 4 shall be less than 500 ohms;
2. LED shall be unlit;
3. Voltage at terminal VDD shall be 12.0 ±1.0 Vdc.

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b. Momentary connection of a wire from terminal VDD to terminal 5 shall cause the PWB assembly to latch in the alarm state such that:

1. Resistance across terminals 2 and 4 shall be greater than 100 K ohms;
2. LED shall be lit.

c. Momentary connection of a wire from terminal VDD to terminal 1 shall be cause return to the conditions of a above.

3.6 Burn-in. Each LAS and PWB assembly when furnished separately shall be subjected to a 24 hour burn-in to reduce the number of infant mortality failures (see 4.5.2.4).

3.7 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.8 Identification marking. The LAS and PWB assembly shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.

3.9 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9, except soldering. These requirements are applicable to wiring, welding, brazing, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsuitable for the purposes intended.

3.9.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, requirement 5.

3.10 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.8) for manufacturing and testing LAS's and PWB assemblies when furnished separately:

Item No.	Description	Identification	Quantity
1.	Stable base drawing of LAS PWB assembly	13220E3031	1

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall be come a part of the contractor's overall inspection

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system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article LAS and PWB assembly when furnished separately shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests.

4.3.2.1 Complete LAS's. Complete LAS's shall be subjected to the tests marked "X" in column 1 of table I. Individual testing of PWB assemblies is not required. Failure of any test shall be cause for rejection.

4.3.2.2 PWB assemblies when furnished separately. PWB assemblies when furnished separately shall be submitted to the tests marked "X" in column 1 of table I, except as noted. Failure of any test shall be cause for rejection.

4.3.2.3 Measurement recording requirements. All measurements, including voltages, currents, resistances, and distances, observed during first article tests shall be recorded on the test data sheets and shall be to the precision necessary to demonstrate compliance with requirements in the appendix test method being conducted.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II. Sampling for tests shall be in accordance with MIL-STD-105 and as specified herein.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major defects specified in 4.5.1. Sample size shall be determined by using MIL-STD-105, table I and table II-A. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.3 Tests.

4.4.3.1 Individual.

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4.4.3.1.1 LAS's. Each LAS shall be subjected to the tests marked "X" in column 2 of table I. Individual testing of PWB assemblies is not required. Failure of any test shall be cause for rejection.

4.4.3.1.2 PWB assemblies when furnished separately. Each PWB assembly when furnished separately shall be subjected to the tests marked "X" in column 2 of table I, except as noted. Failure of any test shall be cause for rejection.

4.4.3.2 Samples.

4.4.3.2.1 LAS's. For normal inspection, five units shall be randomly selected from the monthly production lot. Each sample shall be subjected to the tests marked "X" in column 3 of table I. When tightened inspection is required, sample size shall be eight units. Reduced inspection may be instituted, if approved by responsible Government official, when 10 consecutive monthly sample lots have been subjected to normal inspection, there have been zero defectives on original inspection and production is at a steady rate. Sample size during reduced inspection shall be 3 units. For each sample size, the lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.3.2.2 PWB assemblies when furnished separately. In normal inspection, five units shall be randomly selected from the monthly production lot. Each sample shall be subjected to the tests marked "X" in column 3 of table I. When tightened inspection is required, sample size shall be eight units. Reduced inspection may be instituted, if approved by responsible Government official, when 10 consecutive monthly sample lots have been subjected to normal inspection, there have been zero defectives on original inspection and production is at a steady rate. Sample size during reduced inspection shall be 3 units. For each sample size, the lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.5 Inspection procedure.

4.5.1 Examination. The LAS and PWB assemblies when furnished separately shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatment.
105. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the LAS or PWB assemblies when furnished separately.
106. Any part (or component) not in accordance with the QAP requirements as shown on the drawings.
107. Components missing or not as specified.
108. Assembly not correct.
109. Identification or special marking missing, incomplete or illegible.
110. Interchangeability not as specified.
111. Workmanship including wiring and soldering not as specified.

4.5.2 Tests.

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4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. After successful completion of functional testing, each LAS and PWB assembly when furnished separately shall be subjected to a 24-hour burn-in as specified in 4.5.2.4. Unless otherwise specified herein, the LAS and PWB assemblies when furnished separately shall operate as specified herein without maintenance.

4.5.2.2 Test schedule.

TABLE I. Test schedule.

	Quality Conformance				
First article	Individual	Sample	Test	Req Para	Test Para
1	2	3	4	5	6
X[1/]	X[1/]	-	Functional	3.5.1 to 3.5.3	4.5.2.5.1
X[1/]	X[2/]	-	Functional	3.5.5	4.5.2.5.2
X	X	-	Burn-In	3.6	4.5.2.4
X[1/]	X[1/]	-	Tamper switch tolerance	3.5.1	4.5.2.5.3
X	-	X	Low Temp	3.4	4.5.2.6.1
X	-	X	High Temp	3.4	4.5.2.6.2
X	-	X	Humidity	3.4	4.5.2.6.3
X[1/]	-	-	Shock	3.4.2	4.5.2.6.4
X[1/]	-	-	Durability	3.5.4	4.5.2.7

[1/]Test not required for PWB assemblies when furnished separately.
[2/]Test not required for complete LAS.

4.5.2.3 Test methods. Tests shall be conducted in accordance with the test methods in the appendix and as specified herein. Calibration of the test equipment shall be in accordance with MIL-STD-45662.

4.5.2.4 Burn-in. Each LAS and PWB assembly when furnished separately shall be subjected to a 24-hour burn-in after successful completion of a functional test. During the burn-in, each unit shall be subjected to the function test of appendix test method 1 or 4, as applicable, nine times during the 24-hour period, but no test shall be initiated less than 45 minutes after completion of the previous test period. A complete functional test shall be performed after completion of burn-in. The pre- and post- burn-in tests may be considered a part of the nine times provided they are conducted at the start and end of the burn-in period. Tests involving reversed polarity and current measurement need not be performed during burn-in. Non-compliance with the applicable portions of 3.5 shall constitute a failure. Failures during burn-in shall be analyzed, repaired, and corrective action taken as necessary. When specified in the contract or purchase order (see 6.2), burn-in test reports shall be forwarded to the contracting officer. Any unit failing during burn-in shall be subjected to an additional 24-hour burn-in after failure analysis and repair and successful completion of the function test.

4.5.2.5 Operational tests.

4.5.2.5.1 LAS functional test. The LAS shall be tested in accordance with appendix test method 1. Nonconformance to 3.5.1, 3.5.2, or 3.5.3 shall constitute failure of this test.

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4.5.2.5.2 LAS PWB assembly functional test. The LAS PWB shall be tested in accordance with appendix test method 4. Nonconformance to 3.5.5 shall constitute failure of this test.

4.5.2.5.3 Tamper switch tolerance. The tamper switch shall be tested in accordance with appendix test method 2. Nonconformance to 3.5.1 shall constitute failure of this test.

4.5.2.6 Environmental. The LAS and PWB assemblies when furnished separately shall be subjected to the following series of tests specified in MIL-STD-810. The pretest, performance check during test, and post-test data specified in section 3.2 of MIL-STD-810 shall be obtained by performing the functional test of 4.5.2.5.1 or 4.5.2.5.2 as applicable. The LAS's may not be removed from the environmental chamber during the low and high temperature test; however, when performing the functional test at low and high temperature, the environmental chamber door may be opened long enough to accommodate manual activation of the LAS switches S1, S2, and S3. The total time the door is open shall not exceed one minute per LAS. During the low and high temperature exposure, the LAS enclosure cover shall be off.

4.5.2.6.1 Low temperature. Conduct low temperature test method 502.1, procedure I, storage temperature of -30 °F for not less than 48 hours, equipment operating temperature of -20 °F.

4.5.2.6.2 High temperature. Conduct high temperature test method 501.1 procedure II, except that the storage temperature during step 4 shall be +165 °F and the operating temperature shall be +150 °F.

4.5.2.6.3 Humidity. Conduct test method 507.1, procedure I with the following exceptions:

a. Step 5 shall be for 5 cycles (not less than 120 hours). Perform the functional test at the completion of each cycle.
b. During steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.
c. During step 4, the relative humidity shall be maintained at 92 percent or greater.
d. Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle. The LAS may be removed from the environmental chamber for measurements. However, unsuccessful completion of measurements or unsuccessful return of the LAS to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of the test.

4.5.2.6.4 Shock. Bench handling: Conduct test method 516.2, procedure V.

4.5.2.6.5 Failure criteria. Nonconformance to 3.4 shall constitute failure of these tests.

4.5.2.7 Durability. The LAS shall be tested in accordance with appendix test method 3. Nonconformance to 3.5.4 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Sample size shall be determined by using MIL-STD-105 table I and table II-A. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

112. Preservation method for complete LAS's and PWB assemblies when furnished separately not as specified for levels A, B, and C.
113. Boxed contents not cushioned to prevent movement or damage as specified for levels A or B.
114. Consolidation not as specified for levels A, B or C.
115. Shipping containers not as specified for levels A or B.
116. Quantities packed together in shipping containers exceed gross weight specified for levels A, B or C.
117. Strapping not as specified for levels A or B.
118. Marking missing, illegible, incorrect, or incomplete for levels A, B, or C.
119. Special marking not as specified.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. Each LAS and PWB assembly when furnished separately shall be preserved in accordance with MIL-P-116, submethod IIb, except that, for PWB assemblies when furnished separately, the bag shall be in accordance with MIL-B-117, type 1, class F, style 1. The inner container shall conform to PPP-B-636, type cf, class Domestic, variety SW or DW, grade 200. The outer container shall conform to PPP-B-636, W5c, style optional.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for level A, except submethod IIc shall be used in place of submethod IIb and, for PWB assemblies when furnished separately, the bag shall be in accordance with MIL-B-117, type 1, class F, style 1. Each LAS and PWB assembly when furnished separately shall then be placed in a close-fitting box conforming to PPP-B-636, style optional. The components shall be cushioned within the box to prevent movement and damage.

5.1.3 Level C. Each complete LAS and PWB assembly when furnished separately shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration physical damage during shipment from the supply source to the first using activity. As a minimum, each PWB assembly when furnished separately shall be placed in a bag conforming to MIL-B-117, type 1, class F, style 1, and each LAS and PWB assembly when furnished separately shall be placed in a fiberboard box.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The LAS's and PWB assemblies when furnished separately, preserved as specified in 5.1, shall be packed by the following procedure depending upon procurement of complete LAS's or procurement of PWB assemblies furnished separately.

5.2.1.1 Packing of complete LAS's. Complete LAS's shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional, in quantities not to exceed 200 pounds gross weight.

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Cushioning and blocking within the box used to prevent movement and isolate impact shock shall be in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

5.2.1.2 Packing of PWB assemblies when furnished separately. PWB assemblies when furnished separately, preserved as specified in 5.1, shall be packed together as specified in 5.2.1.1.

5.2.2 Level B. Complete LAS's and PWB assemblies when furnished separately, preserved as specified in 5.1 shall be packed as specified in 5.2.1.1 for level A, except boxes shall be domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, V11C, V13, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the applicable box specification.

5.2.3 Level C. Complete LAS's and PWB assemblies when furnished separately, preserved as specified in 5.1 shall be packed in boxes conforming to PPP-B-636 domestic class. The quantity of the contents shall not exceed the weight or size limitations of the box. Box closure shall be in accordance with the appendix to the box specification.

5.3 Marking. Marking for military levels of protection shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.1.

5.3.1 Special marking. Each shipping container shall be marked with code labels in accordance with MIL-STD-129. The labels shall contain the project code "JID".

6. NOTES

(This section contains information of a general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. The LAS is intended to enable its operator to surreptitiously initiate an alarm when placed under duress; it is a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
c. Nomenclature and quantity of item(s) to be furnished.
d. Technical Data Package List(s), including revision level, to which the item(s) are to be built.
e. When a first article is required for inspection and approval, the number of units required, and the allocation of units required to environmental and durability tests respectively (see 3.2).
f. When Government-loaned property is not required (see 3.10).
g. Level of preservation and packing required (see 5.1 and 5.2).
h. Schedule, content, and format of burn-in failure reports (see 4.5.2.4).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific

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instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such protection or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 Tamper alarm. Alarm resulting from removal of LAS cover.

6.4.2 Duress alarm. Alarm resulting from manual activation of the LAS by security personnel.

6.4.3 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any components; physical change which impairs the mechanical integrity of the LAS; evidence of delamination or water penetration into integrated circuits, printed circuit boards, external components, or physical condition resulting in noncompliance of the item with the provisions of this specification.

6.4.4 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. A QAP indicates the minimum requirements which must be inspected on the product drawings to verify the objectives of the product and assure interchangeability of the repair parts.

6.5 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken then applicable."

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each LAS.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the end items.

6.8 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.10 when applicable (see 6.2).

6.9 Subject term (key word) listing.

Duress switch
Intrusion detection
Physical security

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6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - SH
Air Force - 99

Review activity:
DLA - CS

Preparing activity:
Army - ME

Project 6350-0115

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APPENDIX

TEST METHODS

FOR

SWITCH, ALARM, LATCHING

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether the LAS and PWB assemblies procured under this specification conform to the requirements specified.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

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APPENDIX

TEST METHOD 1

LAS FUNCTIONAL TEST

1. TEST APPARATUS:

a. DC power supply - constant output or adjustable to provide a voltage of 20 ±0.3 Vdc at currents up to 40 mA, load regulation <0.5%, ripple <100 mv p-p, total voltage change due to ambient temperature change and drift with time during the test period <0.5 Vdc.
b. DC voltmeter - between 19 and 21 V, minimum resolution 0.1 Vdc, minimum accuracy 1.5%.
c. DC ammeter - between 10 and 30 mA, minimum resolution 0.1 mA, minimum accuracy 2%, maximum full scale voltage drop (burden voltage) 0.3 V.
d. Ohmmeter - resolution 1% of reading, accuracy 2% of reading.

NOTE: Number of required meters may be minimized by use of multimeter and suitable switching arrangements or by moving test leads as necessary.

2. TEST PROCEDURE:

NOTE: During first article tests all voltage, current, and resistance values required shall be recorded. During production tests, check marks maybe used to denote compliance when the values fall within specified limits.

a. Remove cover.
b. Set power supply for 20.0 ±0.3 Vdc (record voltage) and connect "+" lead through ammeter to terminal TB1-7; connect 20 Vdc return to TB1-8 of the LAS. The alarm Light Emitting Diode (LED) shall not be lit.
c. Observe and record current. It shall be between 10 and 30 mA. After completing measurement, remove ammeter if desired, connecting +20 V lead directly to TB1-7.
d. Connect ohmmeter across terminals TB1-1 and TB1-2.
e. Actuate alarm switch S1 using the actuator plate. The alarm LED shall illuminate and remain on. The duress alarm output impedance read on the ohmmeter shall be more than 100K ohms. Record value.
f. Momentarily depress reset switch S2. The alarm LED shall turn off. The ohmmeter shall read less than 500 ohms. Record value.
g. Disconnect ohmmeter test leads and turn off power supply.
h. Connect ohmmeter between terminals TB1-5 and TB1-6. The tamper alarm output impedance shall be more than 1 megohm.
i. Depress tamper switch S3. The no tamper alarm output impedance shall be less than 1 ohms.
j. Reverse leads into TB1-07 and TB1-8 and turn on power supply. NOTE: Before turning on power supply be sure the LAS enclosure is electrically isolated from the negative power supply terminal.
k. Actuate switch S1, Alarm LED shall not come on.
l. Reverse leads into TB1-7 and TB1-8 (+20 V to TB1-7).
m. Repeat steps d, e, and f to verify no damage has occurred. NOTE: Steps h, i, j. k, l, and m need be performed only during the initial functional test after burn-in (i.e. may be omitted during burn-in and during and after environmental tests).

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APPENDIX

TEST METHOD 2

LAS TAMPER SWITCH TOLERANCE TEST

1. TEST APPARATUS:

a. Ohmmeter - resolution 10% of reading, accuracy 10% of reading.
b. Ruler - Graduated in 1/16-inch increments.

2. TEST PROCEDURE:

a. Connect ohmmeter between terminals TB1-5 and TB1-6. (Install test leads through conduit access hole.)
b. Replace cover.
c. Slowly lift the cover. The ohmmeter shall indicate opening of tamper switch contacts with cover movement between 1/16 and 1/4 inch.

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APPENDIX

TEST METHOD 3

LAS DURABILITY TEST

1. TEST APPARATUS:

a. DC power supply - constant output or adjustable to provide a voltage of 20 ±0.3 Vdc at currents up to 40 mA, load regulation <0.5%, ripple <100 mv p-p, total voltage canmge due to ambient temperature change and drift with time during the test period <0.5 Vdc.
b. DC voltmeter - between 19 and 21 V minimum resolution 0.1 Vdc, minimum accuracy 1.5%.
c. DC ammeter - between 10 and 30 mA, minimum resolution 0.1 mA, minimum accuracy 2%, maximum full scale voltage drop (burden voltage) 0.3 V.
d. Ohmmeter - resolution 1 % of reading, accuracy 2% of reading.

2. TEST PROCEDURE:

a. Perform test method 1.
b. Manually or mechanically activate switch S1, using the actuator.
c. Manually or mechanically activate reset switch S2.
d. Repeat steps b. and c. 1500 times each.
e. Repeat test method 1.

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APPENDIX

TEST METHOD 4

LAS PRINTED WIRING BOARD ASSEMBLY FUNCTIONAL TEST

1. TEST APPARATUS:

a. DC power supply - constant output or adjustable to provide a voltage of 20 ±0.3 Vdc at currents up to 40 mA, load regulation <0.5%, ripple < 100 mv p-p, total voltage change due to ambient temperature change and drift with time during the test period <0.5 Vdc.
b. DC Voltmeter - between 19 and 21 V, minimum resolution 0.1 Vdc, minimum accuracy 1.5%.
c. DC Ammeter - Between 10 and 30 mA, minimum resolution 0.1 mA, minimum accuracy 2%, maximum full scale voltage drop (burden voltage) 0.3 V.
d. Ohmmeter - resolution 1% of reading, accuracy 2% of reading.
e. LED, drawing D13220E3816 (contained in TA13220E3800).

NOTE: Number of required meters may be minimized by use of suitable switching arrangements or by moving test leads as necessary.

2. TEST PROCEDURE:

NOTE: During first article tests all voltage, current, and resistance values required shall be recorded. During production tests, check marks may be used to denote compliance when the values fall within specified limits.

a. Connect the LED (item 1e) with its anode to terminal "VDD" and cathode to terminal 6.
b. Set power supply for 20.0 ±0.3 Vdc and connect positive lead through dc ammeter to terminal "+20"; connect negative power supply lead to terminal 3. The LED shall not be lit.
c. Observe and record current. It shall be between 10 and 30 mA. After completing measurement, remove ammeter if desired, connecting +20 V lead directly to terminal "+20".
d. Measure voltage at terminal "VDD". It shall be 12.0 ±1.0 Vdc.
e. Connect ohmmeter across terminals 2 and 4.
f. Momentarily connect jumper between terminal "VDD" and terminal 5. The LED shall illuminate and remain on. The ohmmeter shall read more than 100K ohms. Record value.
g. Momentarily connect jumper between terminal 1. The LED shall turn off and remain off after removing jumper. The ohmmeter shall read less tan 500 ohms. Record value.
h. Disconnect ohmmeter test leads and turn off power supply.
i. Connect positive lead of the power supply to terminal 3 and the negative lead to terminal "+20" and turn on power supply.
j. Momentarily connect jumper between terminal "VDD" and terminal 5. The LED shall not illuminate.
k. Turn off power supply, reverse power supply leads into terminals "+20" and 3, then turn on power supply.
l. Repeat steps e, f, and g to verify no damage has occurred.

NOTE: Steps i, j, and k need to be performed only during the initial functional test after burn-in (i.e. may be omitted during burn-in and during and after environmental tests).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-S-52860D	**2. DOCUMENT DATE (YYMMDD)** 901206

3. DOCUMENT TITLE
SWITCH, ALARM, LATCHING, SA-1954/FSS-9(V)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME BELVOIR CECOM SOUTH	b. TELEPHONE *Include Area Code)* (1) Commercial (703) 704-3468 (2) AUTOVON 654-3468
c. ADDRESS *(Include Zip Code)* 10115 GRIDLEY ROAD SUITE 228 ATTN: AMSEL-LC-IEW-D-ED FORT BELVOIR, VA 22060-5849	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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